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# "Development of AI-Driven Social Platform with FAQ Support "

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Pitambar Yadav

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Description

This work presents an AI-assisted social and academic support chatbot integrated into a web-based social platform for Sri Ramakrishna Engineering College students (SREC Social Bot). The system combines secure email-based login with role-based access (student/admin), a college-style news feed, and an embedded chatbot that answers SREC-specific queries, guides study planning, and restricts access to sensitive resources like question papers to authenticated users only. The backend is built with Flask and Firebase Realtime Database for posts, likes, comments, and role management, while a lightweight local LLM and Google Custom Search API handle conversational replies and SREC-website search. Experimentation with real-like student queries showed that rule-based intent handling plus web search gives more accurate, trustworthy answers for college information than a pure LLM approach, while latency remains low enough for interactive use. The findings suggest that a hybrid design—rules + search + small LLM—can provide a practical, low-cost AI assistant for campus information, mental-health friendly responses, and access-controlled academic resources, suitable for deployment and further extension in Indian college environments.

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Page 1

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## 20CS295 – Mini Project I / II / Project Work

# "Development of AI-Driven Social Platform with FAQ Support "

Type of Project: Application

Expected Project Outcome : Product


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Date : 11.10.2025





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- In many colleges, event updates and announcements are conveyed manually via students/class Representatives.
- Social media is not specifically built for college needs—thus missing relevant academic features.....
- There is no dedicated platform that provides access to past year question papers and academic materials in an organized way. Our chatbot has been enhanced to share links to past year question papers upon request instantly.
- Some institutions use Learning Management Systems (LMS) like Moodle or Google Classroom, but they do not integrate chatbot-based FAQs, social interaction, or student analytics in one platform.
- Our project addresses these gaps through a unified web platform with Firebase, an AI-powered FAQ Chatbot, and an NLP-based text analyzer.





## Problem identification, scope and objectives (Contd.,)

**Problem Definition**

- Students lack a **dedicated, secure, and interactive college platform** for communication, event updates, academic resources, and FAQs. Existing solutions (social media groups, notice boards) are fragmented, unreliable, and not tailored to college needs.

**Scope**

A secure, college-only platform for SREC where students can post content, view club events, access past papers, and get real-time help through an AI chatbot — all in one place to boost academic and social engagement.

## Objectives (Max 3)

- Enable students to post content, view club events, and connect socially, with a sentiment analyzer to understand post tone.
- Build a secure, centralized platform for all SREC students — especially newcomers — to access academic and campus information easily.
- Provide quick access to past year question papers and real-time help through an AI-powered FAQ chatbot.





## Literature Survey – 01

| Experiment | Inference | Problem gap |
|---|---|---|
| Development of a conversational AI bot prototype designed to support efficient learning by providing intelligent, interactive, and personalized assistance to learners. | The study showed that the AI bot boosts learning by increasing engagement, giving timely guidance, and supporting adaptive pathways, proving its value as an educational tool. | For further work, it is suggested to use different geometries with the modification methods that are associated with the increasing evaporation surface area, such as fins and cylinders. |

- **Source:** Ahmed Elragal, Ali Ismail Awad, Ingemar Andersson, and Jörgen Nilsson, “A Conversational AI Bot for Efficient Learning: A Prototypical Design”, IEEE Access, vol. 12, 2024, DOI: 10.1109/ACCESS.2024.3476953.

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| Experiment | Inference | Problem gap |
|---|---|---|
| Developed and deployed MoodleBot, an LLM-based chatbot using a Retrieval-Augmented Generation (RAG) architecture integrated into Moodle LMS, and evaluated it using the Technology Acceptance Model (TAM) and accuracy metrics. | Achieved high perceived usefulness and ease of use, with 88% accuracy in answering course-related queries, effectively supporting self-regulated learning. | The solution remains effective but faces issues like hallucinations, bias, scalability, and missing a social-style platform for student interaction, which CampusConnect addresses. |

- **Source:** Neumann, A. T., Yin, Y., Sowe, S., Decker, S., & Jarke, M. (2025). An LLM-Driven Chatbot in Higher Education for Databases and Information Systems. *IEEE Trans. Educ.*, 68(1). https://doi.org/10.1109/TE.2024.3467912



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